

Decidable Properties of Context-Free Languages

Membership Question:

for context-free grammar G
find if string $w \in L(G)$

Membership Algorithms: Parsers

- Exhaustive search parser
- **CYK** parsing algorithm

Empty Language Question:

for context-free grammar G
find if $L(G) = \emptyset$

Algorithm:

1. Remove useless variables
2. Check if start variable S is useless

Infinite Language Question:

for context-free grammar G
find if $L(G)$ is infinite

Algorithm:

1. Remove useless variables
2. Remove unit and λ productions
3. Create dependency graph for variables
4. If there is a loop in the dependency graph then the language is infinite

Example: $S \rightarrow AB$

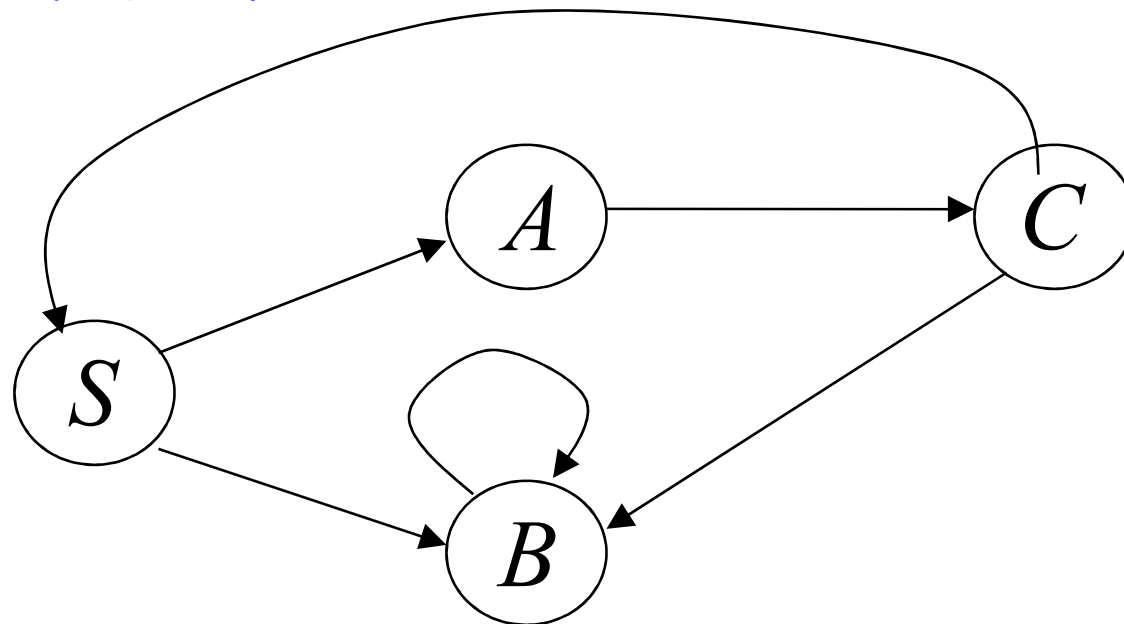
$A \rightarrow aCb \mid a$

$B \rightarrow bB \mid bb$

$C \rightarrow cBS$

Dependency graph

Infinite language



$$S \rightarrow AB$$

$$A \rightarrow aCb \mid a$$

$$B \rightarrow bB \mid bb$$

$$C \rightarrow cBS$$

$$S \Rightarrow AB \Rightarrow aCbB \Rightarrow acBSbB \Rightarrow acbbSb bb$$

$$\overset{*}{S \Rightarrow acbbSb bb} \overset{*}{\Rightarrow (acbb)^2 S (bbb)^2}$$

$$\overset{*}{\Rightarrow (acbb)^i S (bbb)^i}$$

The Pumping Lemma for Context-Free Languages

Take an **infinite** context-free language



Generates an infinite number
of different strings

Example:

$$S \rightarrow AB$$

$$A \rightarrow aBb$$

$$B \rightarrow Sb$$

$$B \rightarrow b$$

$$S \rightarrow AB$$

$$A \rightarrow aBb$$

$$B \rightarrow Sb$$

$$B \rightarrow b$$

A derivation:

Variables are repeated

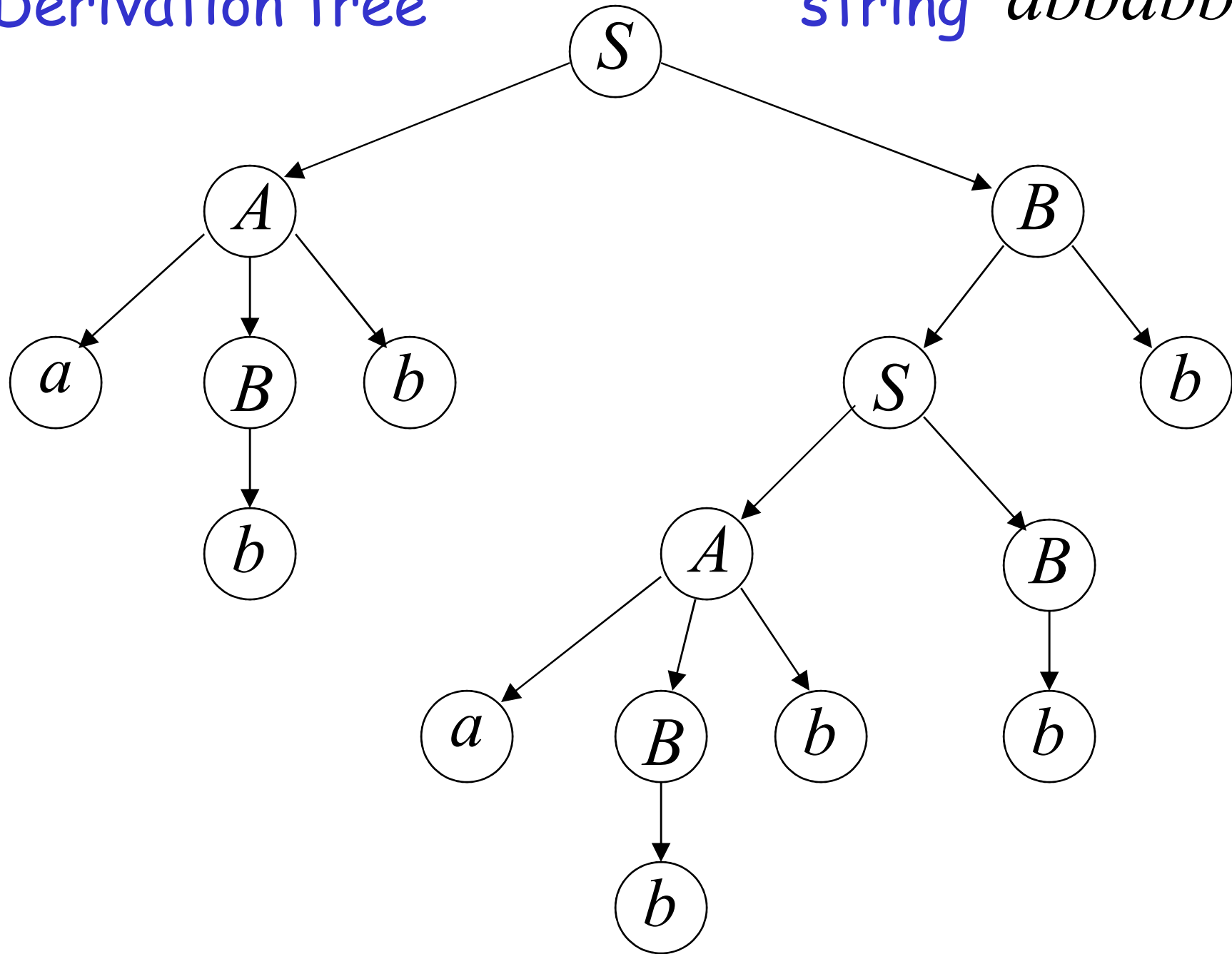
$$S \Rightarrow AB \Rightarrow aBbB \Rightarrow abb\underline{B} \Rightarrow$$

$$\Rightarrow abbSb \Rightarrow abbABb \Rightarrow abbaBbBb \Rightarrow$$

$$\Rightarrow abbabb\underline{B}b \Rightarrow abbabbbb$$

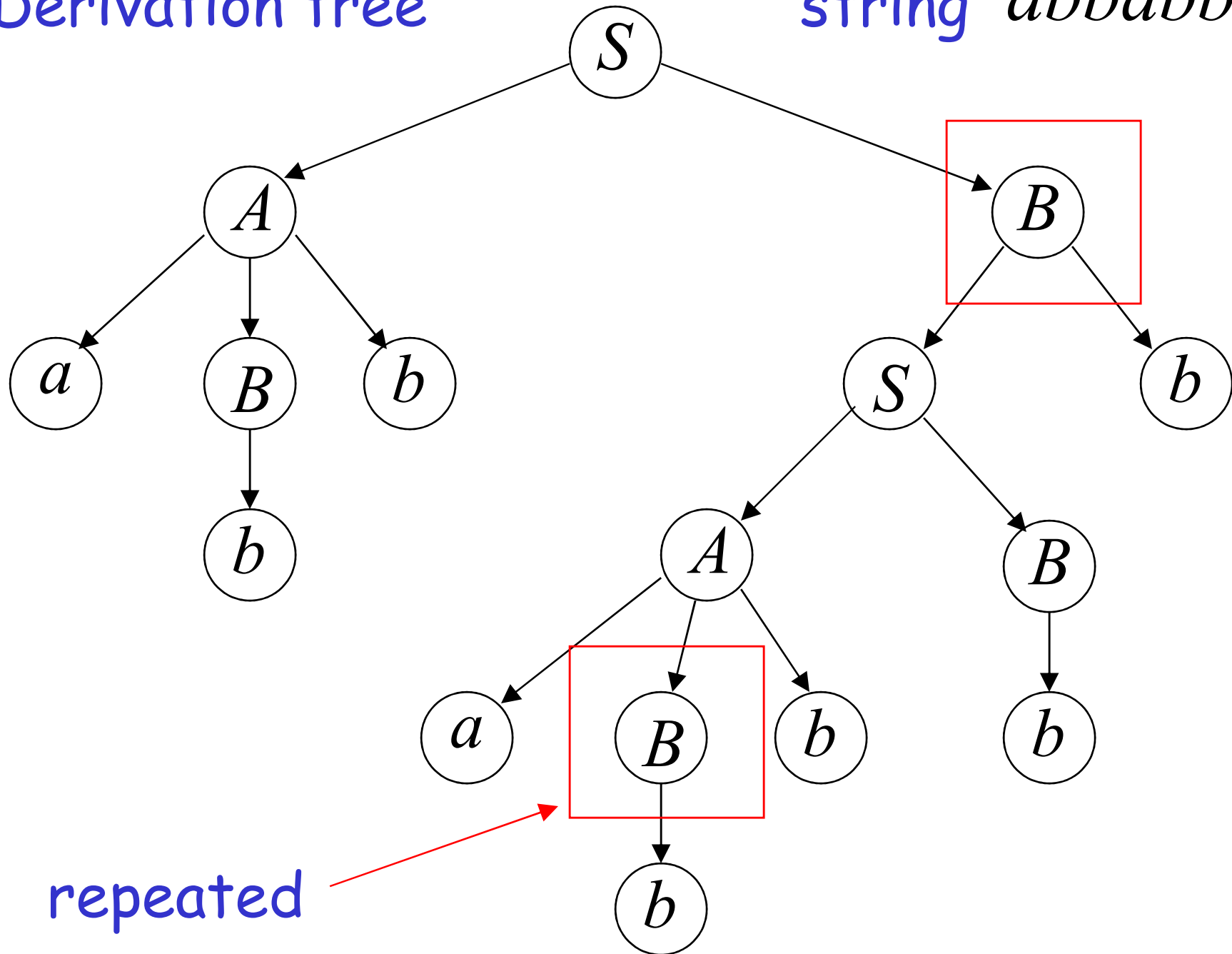
Derivation tree

string *abbabbbb*

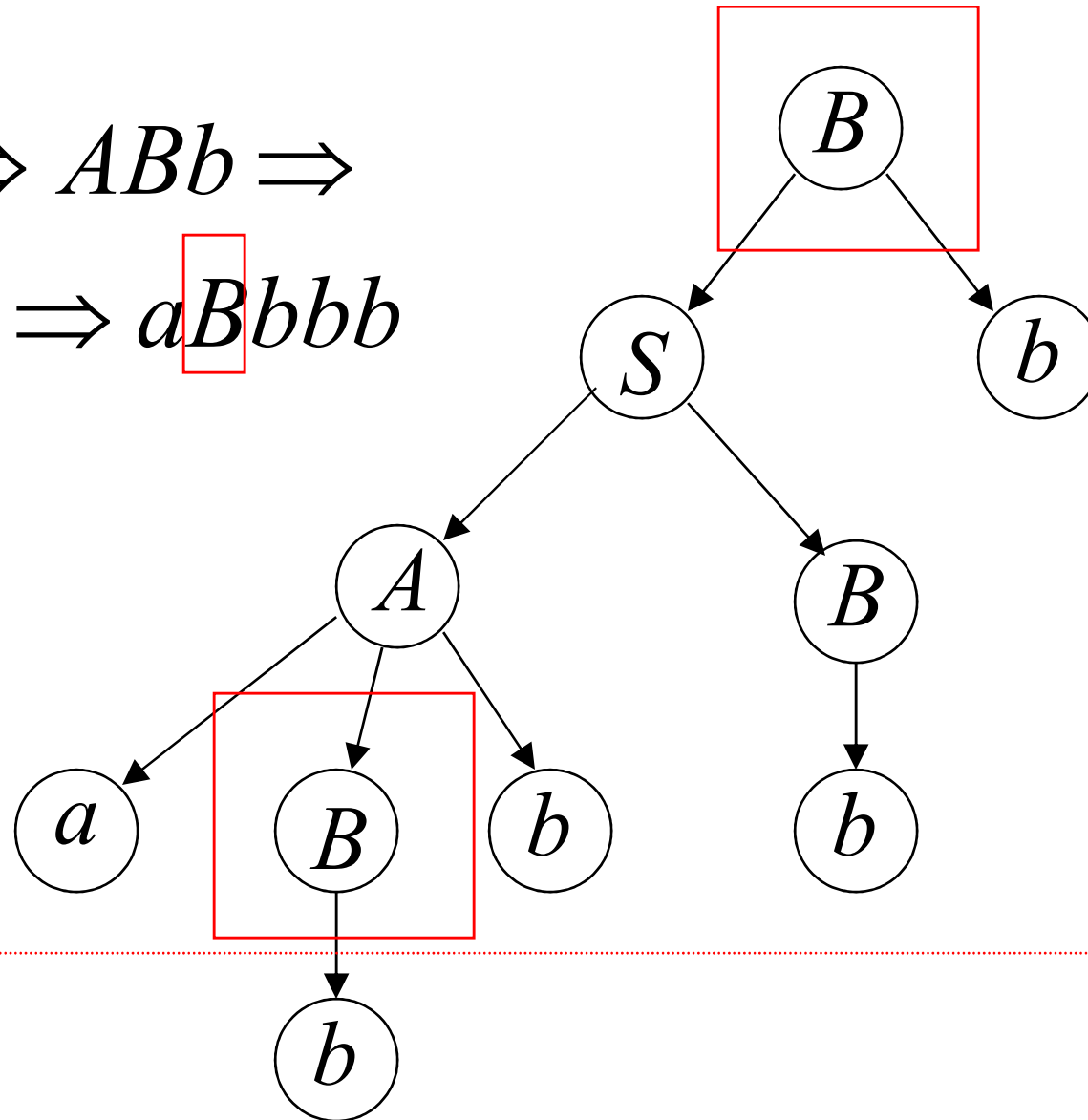


Derivation tree

string *abbabbbb*

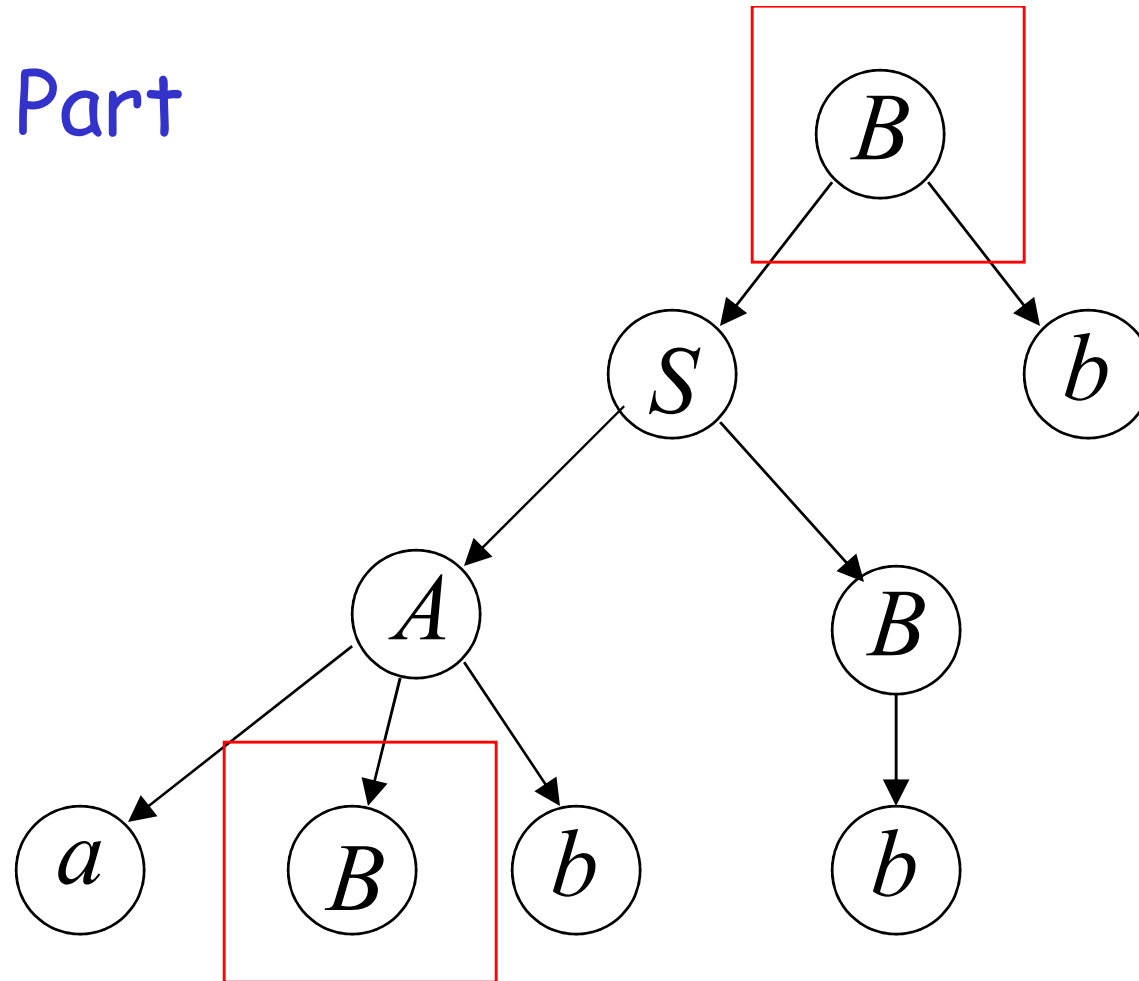


$B \Rightarrow Sb \Rightarrow ABb \Rightarrow$
 $\Rightarrow aBbBb \Rightarrow aBbbb$



$B \Rightarrow b$

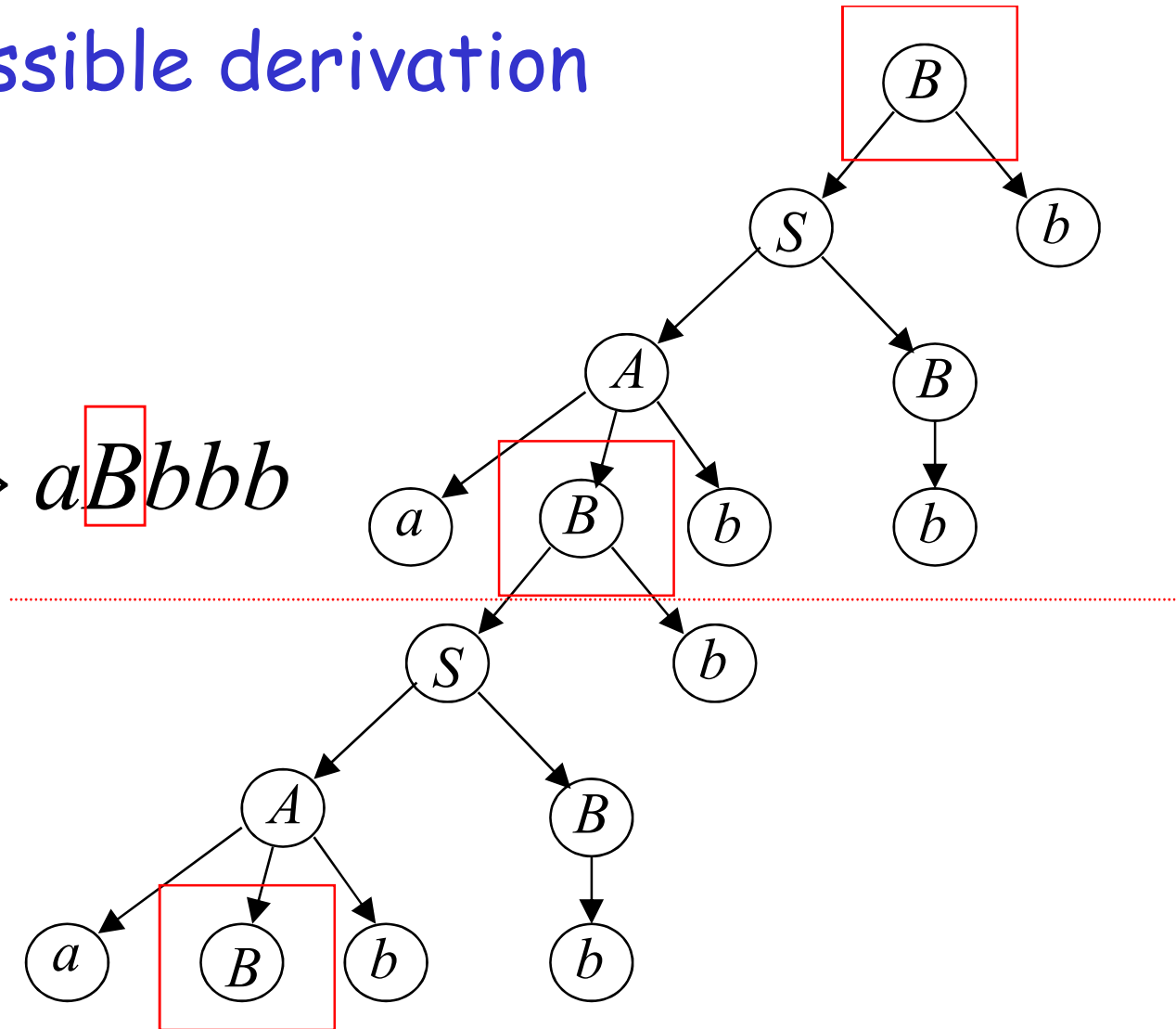
Repeated Part



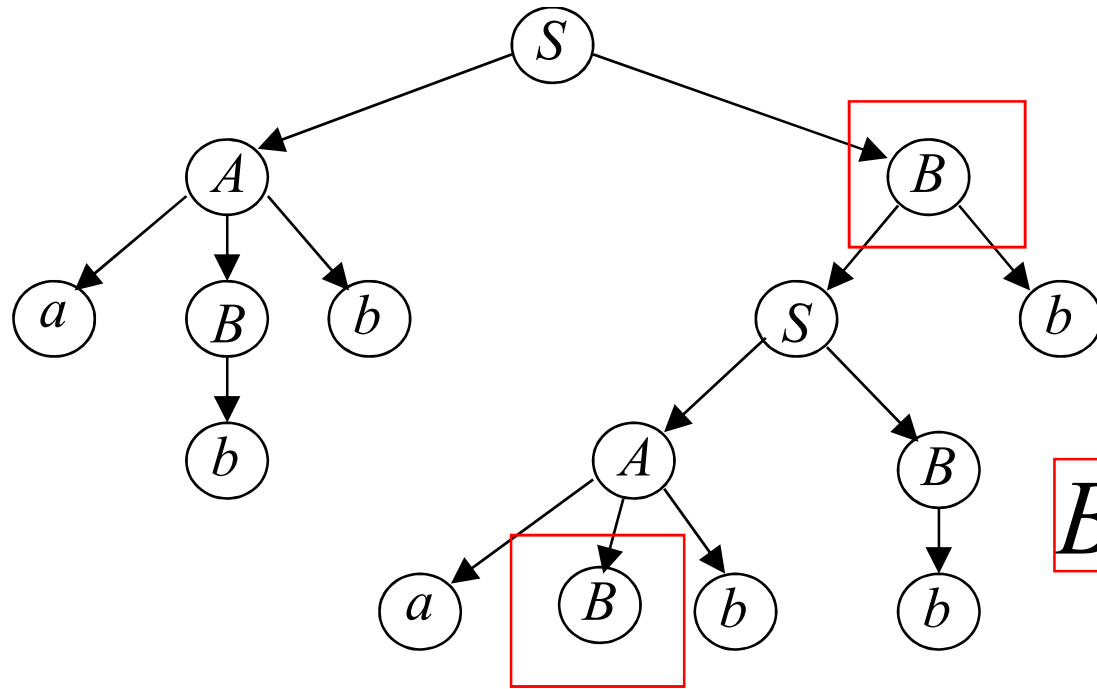
$$\boxed{B} \Rightarrow \dots \Rightarrow a\boxed{B}bbbb$$

Another possible derivation

$$\boxed{B} \Rightarrow \dots \Rightarrow a\boxed{B}bbb$$

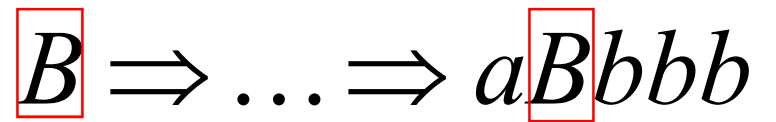


$$\boxed{B} \Rightarrow \dots \Rightarrow a\boxed{B}bbb \dots \Rightarrow aa\boxed{B}bbbbb$$

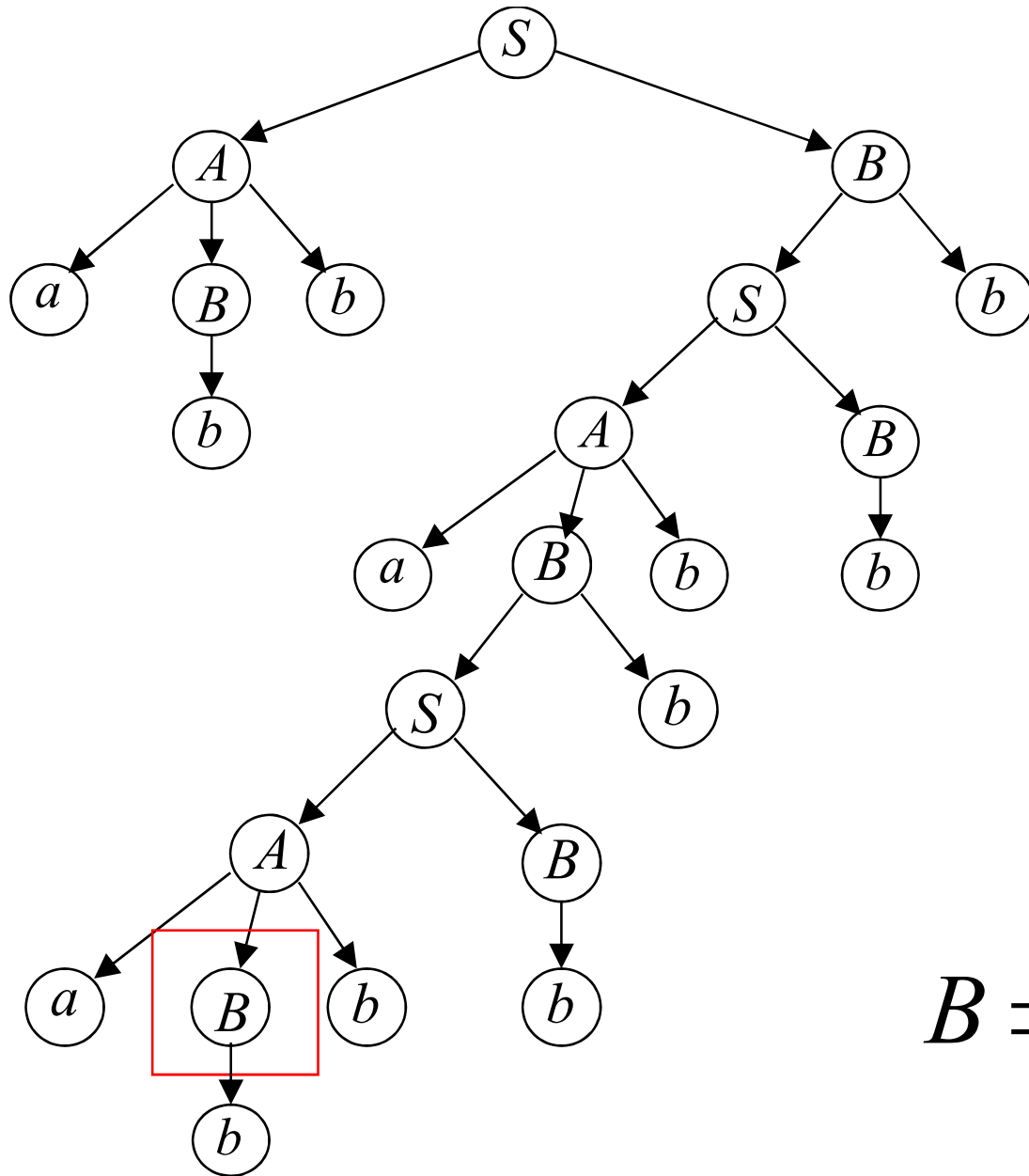


$\boxed{B} \Rightarrow \dots \Rightarrow a\boxed{B}bbbb$

$S \Rightarrow \dots \Rightarrow abba\boxed{B}bbbb$



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$B \Rightarrow b$

$S \Rightarrow \dots \Rightarrow abbaa\boxed{B}bbbbbb \Rightarrow abbaa\boxed{b}bbbbbb$

$$S \Rightarrow \dots \Rightarrow abbaabbbbbbb$$

Therefore, the string

abbaabbbbbbb

is also generated by the grammar

We know: $B \Rightarrow b$

$B \Rightarrow \dots \Rightarrow aBbbb$

$S \Rightarrow \dots \Rightarrow abbaBbbb$

We also know this string is generated:

$S \Rightarrow \dots \Rightarrow abba\boxed{B}bbb \Rightarrow$
 $\Rightarrow abbaabbbb$

We know: $B \Rightarrow b$

$B \Rightarrow \dots \Rightarrow aBbbb$

$S \Rightarrow \dots \Rightarrow abbaBbbb$

Therefore, this string is also generated:

$S \Rightarrow \dots \Rightarrow abbaBbbb \Rightarrow$
 $\Rightarrow abbaaBbbbbbb \Rightarrow$
 $\Rightarrow abbaabbbbbbb$

We know: $B \Rightarrow b$

$$B \Rightarrow \dots \Rightarrow aBbbb$$

$$S \Rightarrow \dots \Rightarrow abbaBbbb$$

Therefore, this string is also generated:

$$\begin{aligned} S &\Rightarrow \dots \Rightarrow abbaBbbb \Rightarrow \\ &\Rightarrow abba(a)B(bbb)bbb \\ &\Rightarrow abba(a)^2 B(bbb)^2 bbb \\ &\Rightarrow abba(a)^2 b(bbb)^2 bbb \end{aligned}$$

We know: $B \Rightarrow b$

$$B \Rightarrow \dots \Rightarrow aBbbb$$

$$S \Rightarrow \dots \Rightarrow abbaBbbb$$

Therefore, this string is also generated:

$$S \Rightarrow \dots \Rightarrow abbaBbbb \Rightarrow$$

$$\Rightarrow \dots$$

$$\Rightarrow abba(a)^i B(bbb)^i bbb$$

$$\Rightarrow abba(a)^i b(bbb)^i bbb$$

Therefore, knowing that

abbabbbb

is generated by grammar G ,
we also know that

$abba(a)^i b(bbb)^i bbb$

is generated by G

In general:

We are given an infinite
context-free grammar G

Assume G has no unit-productions
no λ -productions

Take a string $w \in L(G)$
with length bigger than

m  (Number of productions) 
(Largest right side of a production)

Consequence:

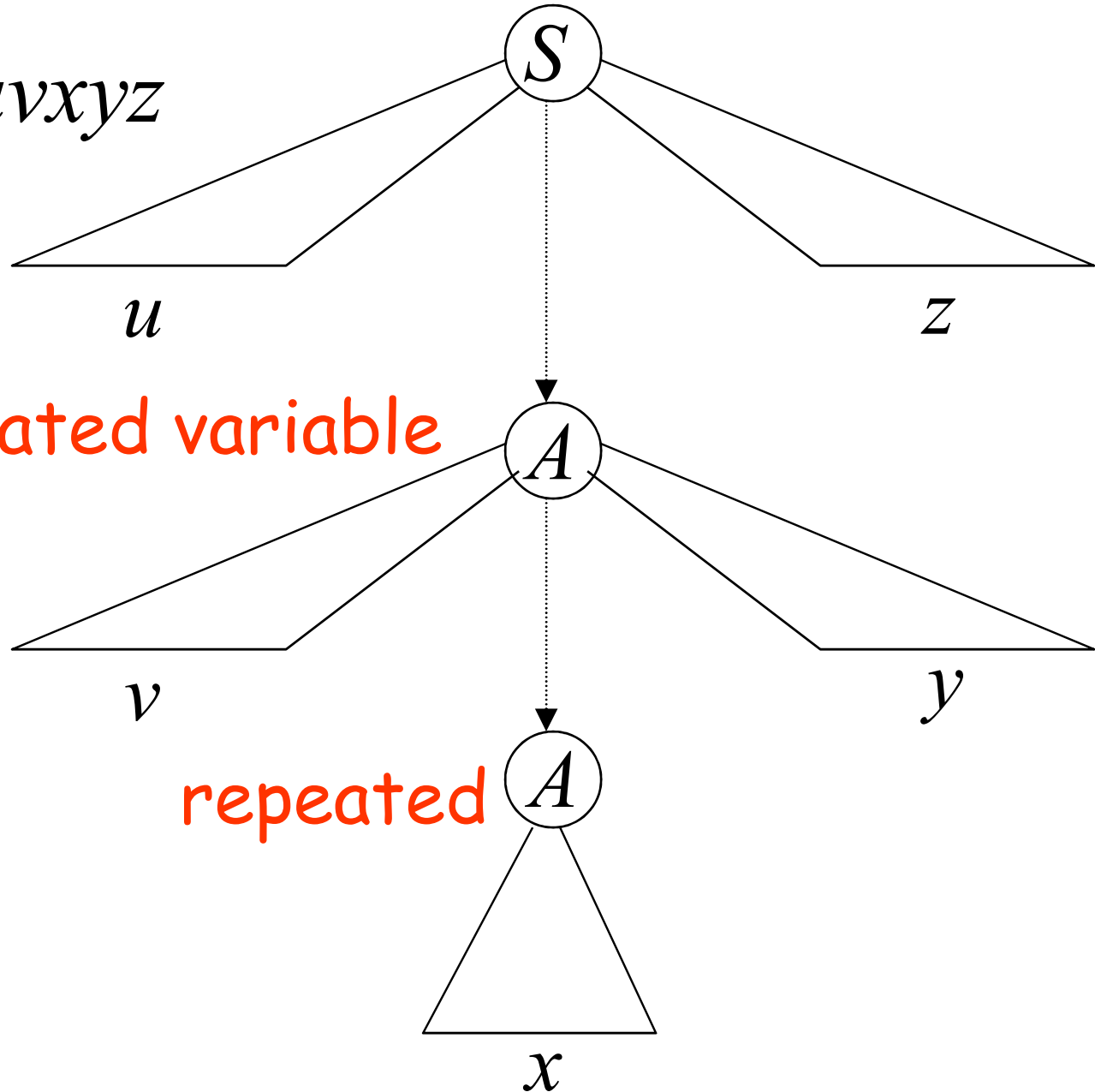
Some variable must be repeated
in the derivation of w

u, v, x, y, z : strings of terminals

String $w = uvxyz$

Last repeated variable

repeated

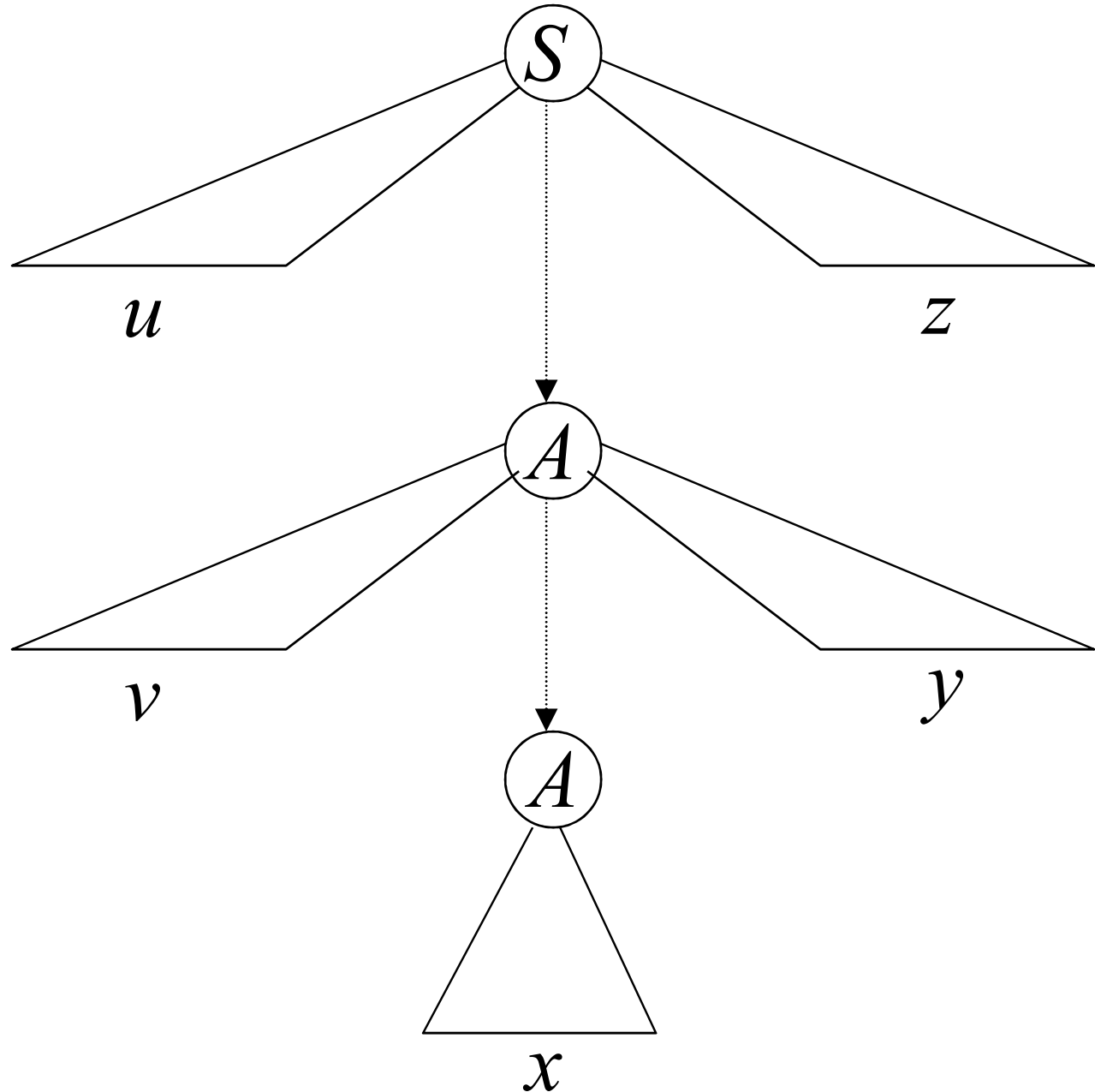


Possible
derivations:

$$S \xRightarrow{*} uAz$$

$$A \xRightarrow{*} vAy$$

$$A \xRightarrow{*} x$$



We know:

$$S \overset{*}{\Rightarrow} uAz$$

$$A \overset{*}{\Rightarrow} vAy$$

$$A \overset{*}{\Rightarrow} x$$

This string is also generated:

$$S \overset{*}{\Rightarrow} uAz \overset{*}{\Rightarrow} uxz$$

$$uv^0xy^0z$$

We know:

$$S \overset{*}{\Rightarrow} uAz$$

$$A \overset{*}{\Rightarrow} vAy$$

$$A \overset{*}{\Rightarrow} x$$

This string is also generated:

$$S \overset{*}{\Rightarrow} uAz \overset{*}{\Rightarrow} uvAyz \overset{*}{\Rightarrow} uvxyz$$

The original $w = uv^1xy^1z$

We know:

$$S \overset{*}{\Rightarrow} uAz$$

$$A \overset{*}{\Rightarrow} vAy$$

$$A \overset{*}{\Rightarrow} x$$

This string is also generated:

$$S \overset{*}{\Rightarrow} uAz \overset{*}{\Rightarrow} uvAyz \overset{*}{\Rightarrow} uvvAyyz \overset{*}{\Rightarrow} uvvxyyz$$

$$uv^2xy^2z$$

We know:

$$S \xRightarrow{*} uAz$$

$$A \xRightarrow{*} vAy$$

$$A \xRightarrow{*} x$$

This string is also generated:

$$S \xRightarrow{*} uAz \Rightarrow uvAyz \xRightarrow{*} uvvAyyz \Rightarrow$$

$$\xRightarrow{*} uvvvAyyyyz \xRightarrow{*} uvvvxyyyz$$

$$uv^3xy^3z$$

We know:

$$S \xRightarrow{*} uAz$$

$$A \xRightarrow{*} vAy$$

$$A \xRightarrow{*} x$$

This string is also generated:

$$\begin{aligned} S &\xRightarrow{*} uAz \xRightarrow{*} uvAyz \xRightarrow{*} uvvAyyz \xRightarrow{*} \\ &\xRightarrow{*} uvvvAyyyzyz \xRightarrow{*} \dots \\ &\xRightarrow{*} uvvv \dots vAy \dots yyyz \xRightarrow{*} \\ &\xRightarrow{*} uvvv \dots vxy \dots yyyz \end{aligned}$$

$$uv^i xy^i z$$

Therefore, any string of the form

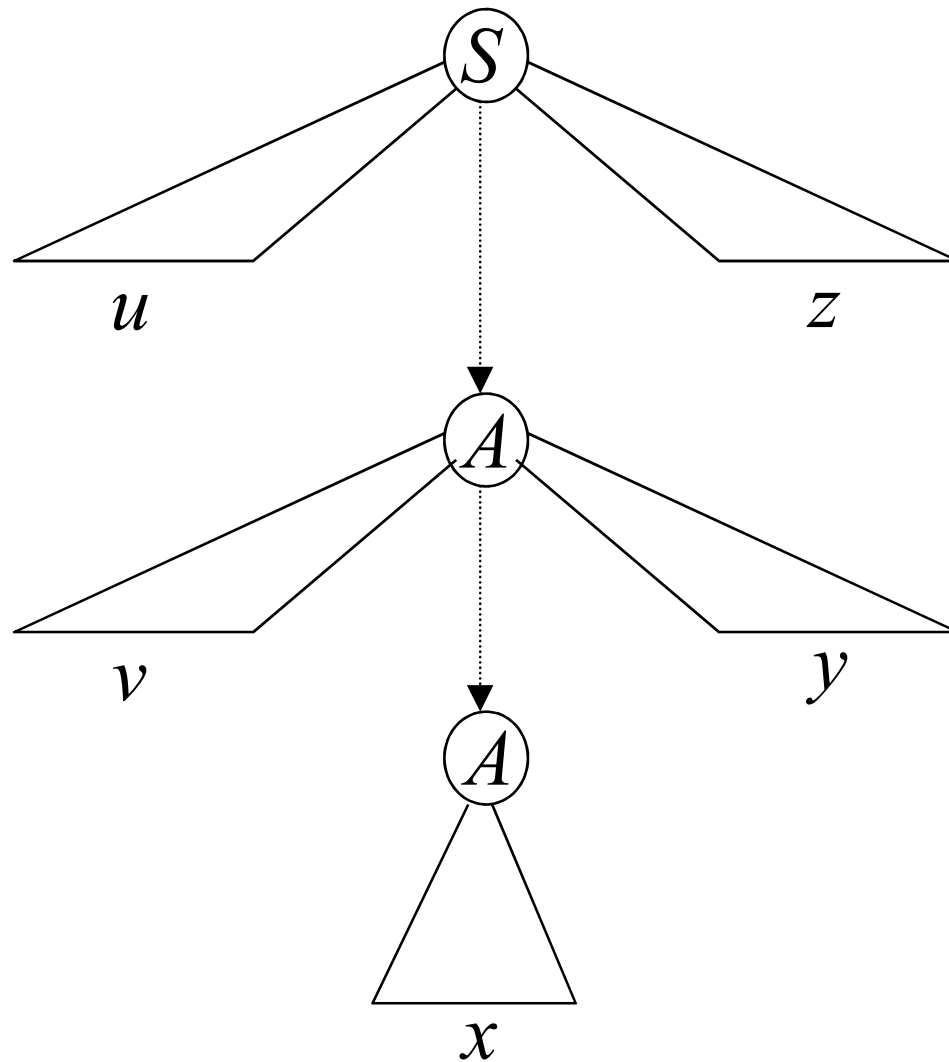
$$uv^i xy^i z \qquad i \geq 0$$

is generated by the grammar G

Therefore,

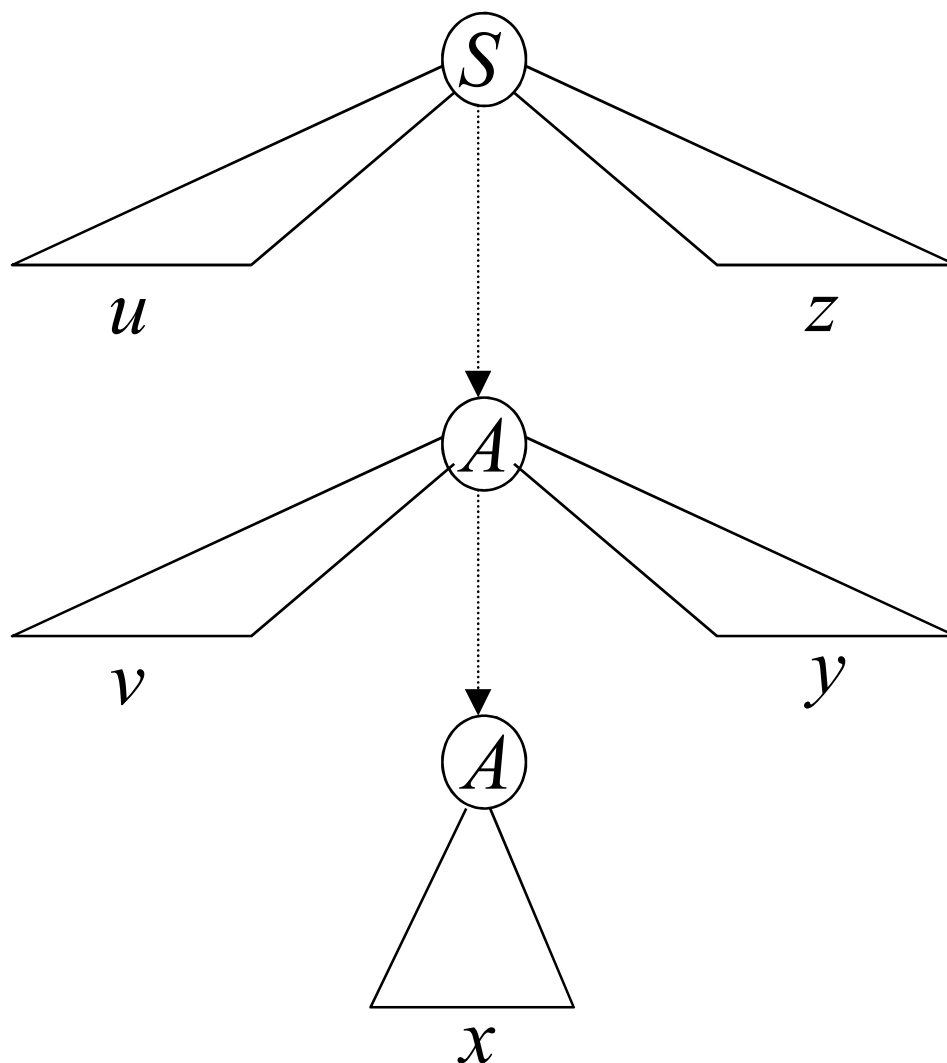
knowing that $uvxyz \in L(G)$

we also know that $uv^i xy^i z \in L(G)$



Observation: $|vxy| \leq m$

Since A is the last repeated variable



Observation: $|vy| \geq 1$

Since there are no unit or λ productions

The Pumping Lemma:

For infinite context-free language L

there exists an integer m such that

for any string $w \in L$, $|w| \geq m$

we can write $w = uvxyz$

with lengths $|vxy| \leq m$ and $|vy| \geq 1$

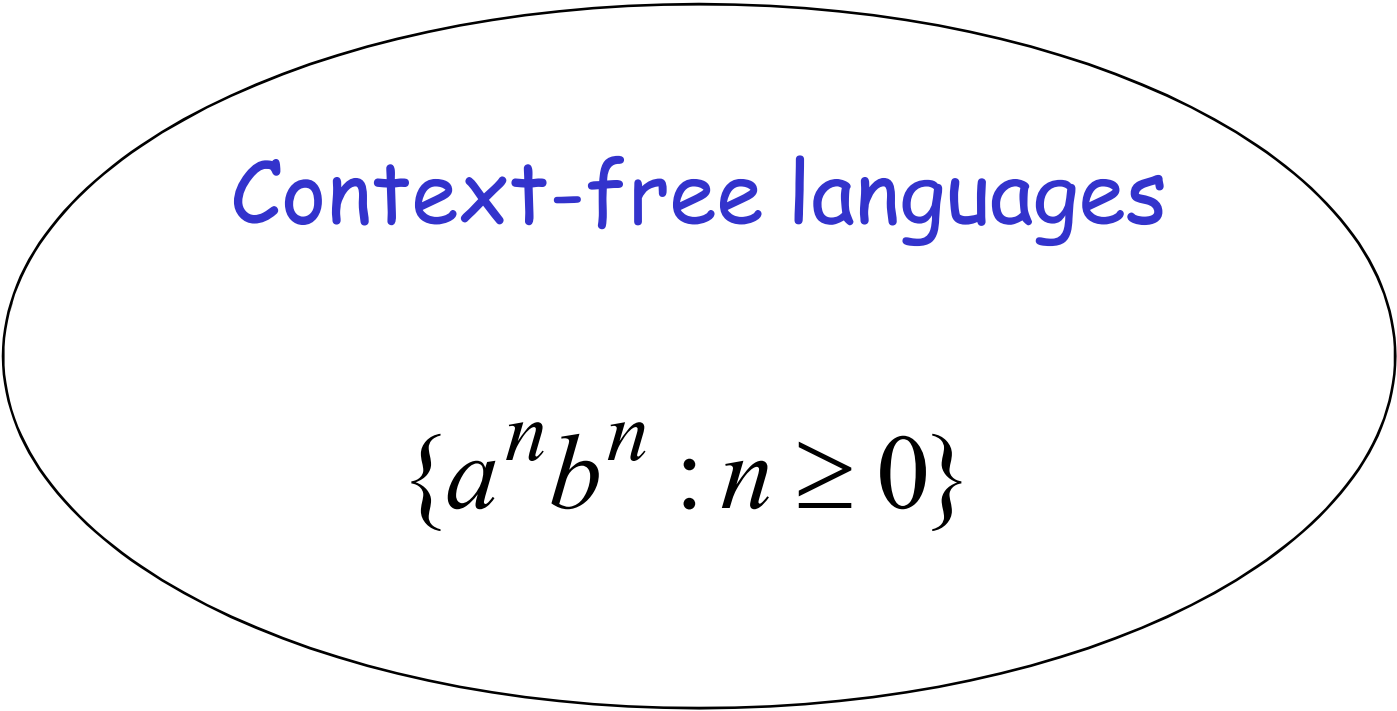
and it must be:

$$uv^i xy^i z \in L, \quad \text{for all } i \geq 0$$

Applications of The Pumping Lemma

Non-context free languages

$$\{a^n b^n c^n : n \geq 0\}$$



Context-free languages

$$\{a^n b^n : n \geq 0\}$$

Theorem: The language

$$L = \{a^n b^n c^n : n \geq 0\}$$

is **not** context free

Proof: Use the Pumping Lemma
for context-free languages

$$L = \{a^n b^n c^n : n \geq 0\}$$

Assume for contradiction that L
is context-free

Since L is context-free and infinite
we can apply the pumping lemma

$$L = \{a^n b^n c^n : n \geq 0\}$$

Pumping Lemma gives a magic number m
such that:

Pick any string $w \in L$ with length $|w| \geq m$

We pick: $w = a^m b^m c^m$

$$L = \{a^n b^n c^n : n \geq 0\}$$

$$w = a^m b^m c^m$$

We can write: $w = uvxyz$

with lengths $|vxy| \leq m$ and $|vy| \geq 1$

$$L = \{a^n b^n c^n : n \geq 0\}$$

$$w = a^m b^m c^m$$

$$w = uvxyz \quad |vxy| \leq m \quad |vy| \geq 1$$

Pumping Lemma says:

$$uv^i xy^i z \in L \quad \text{for all} \quad i \geq 0$$

$$L = \{a^n b^n c^n : n \geq 0\}$$

$$w = a^m b^m c^m$$

$$w = uvxyz \quad |vxy| \leq m \quad |vy| \geq 1$$

We examine all the possible locations
of string vxy in w

$$L = \{a^n b^n c^n : n \geq 0\}$$

$$w = a^m b^m c^m$$

$$w = uvxyz \quad |vxy| \leq m \quad |vy| \geq 1$$

Case 1: vxy is within a^m

$$\overbrace{aaa...aaa}^m \overbrace{bbb...bbb}^m \overbrace{ccc...ccc}^m$$

$u \quad vxy \quad z$

$$L = \{a^n b^n c^n : n \geq 0\}$$

$$w = a^m b^m c^m$$

$$w = uvxyz \quad |vxy| \leq m \quad |vy| \geq 1$$

Case 1: v and y consist from only a

$$\begin{array}{c} \overbrace{aaa \dots aaa}^m \quad \overbrace{bbb \dots bbb}^m \quad \overbrace{ccc \dots ccc}^m \\ \underbrace{aaa \dots aaa}_{u \quad vxy} \quad \underbrace{bbb \dots bbb}_{z} \end{array}$$

$$L = \{a^n b^n c^n : n \geq 0\}$$

$$w = a^m b^m c^m$$

$$w = uvxyz \quad |vxy| \leq m \quad |vy| \geq 1$$

Case 1: Repeating v and y

$$k \geq 1$$

$$\overbrace{aaaaaa \dots aaaaaa}^{m+k} \overbrace{bbb \dots bbb}^m \overbrace{ccc \dots ccc}^m$$

$$\underbrace{\quad}_u \underbrace{\quad}_{v^2 xy^2} \underbrace{\quad}_z$$

$$L = \{a^n b^n c^n : n \geq 0\}$$

$$w = a^m b^m c^m$$

$$w = uvxyz \quad |vxy| \leq m \quad |vy| \geq 1$$

Case 1: From Pumping Lemma: $uv^2xy^2z \in L$

$$k \geq 1$$

$$\begin{array}{c}
 \overbrace{aaaaaa \dots aaaaaa}^{m+k} \quad \overbrace{bbb \dots bbb}^m \quad \overbrace{ccc \dots ccc}^m \\
 \underbrace{\quad \quad \quad}_u \quad \underbrace{\quad \quad \quad}_{v^2xy^2} \quad \underbrace{\quad \quad \quad}_z
 \end{array}$$

$$L = \{a^n b^n c^n : n \geq 0\}$$

$$w = a^m b^m c^m$$

$$w = uvxyz \quad |vxy| \leq m \quad |vy| \geq 1$$

Case 1: From Pumping Lemma: $uv^2xy^2z \in L$
 $k \geq 1$

However: $uv^2xy^2z = a^{m+k}b^m c^m \notin L$

Contradiction!!!

$$L = \{a^n b^n c^n : n \geq 0\}$$

$$w = a^m b^m c^m$$

$$w = uvxyz \quad |vxy| \leq m \quad |vy| \geq 1$$

Case 2: vxy is within b^m

$$\begin{array}{c} \overbrace{aaa \dots aaa}^m \quad \overbrace{bbb \dots bbb}^m \quad \overbrace{ccc \dots ccc}^m \\ \underbrace{\hspace{1.5cm}}_u \quad \underbrace{\hspace{1.5cm}}_{vxy} \quad \underbrace{\hspace{1.5cm}}_z \end{array}$$

$$L = \{a^n b^n c^n : n \geq 0\}$$

$$w = a^m b^m c^m$$

$$w = uvxyz \quad |vxy| \leq m \quad |vy| \geq 1$$

Case 2: Similar analysis with case 1

$$\begin{array}{c}
 \overbrace{aaa \dots aaa}^m \quad \overbrace{bbb \dots bbb}^m \quad \overbrace{ccc \dots ccc}^m \\
 \underbrace{\hspace{1.5cm}}_u \quad \underbrace{\hspace{1.5cm}}_{vxy} \quad \underbrace{\hspace{1.5cm}}_z
 \end{array}$$

$$L = \{a^n b^n c^n : n \geq 0\}$$

$$w = a^m b^m c^m$$

$$w = uvxyz \quad |vxy| \leq m \quad |vy| \geq 1$$

Case 3: vxy is within c^m

$$\begin{array}{c} \overbrace{aaa \dots aaa}^m \quad \overbrace{bbb \dots bbb}^m \quad \overbrace{ccc \dots ccc}^m \\ \underbrace{aaa \dots aaa \quad bbb \dots bbb}_{u} \quad \underbrace{ccc \dots ccc}_{\begin{array}{c} vxy \quad z \end{array}} \end{array}$$

$$L = \{a^n b^n c^n : n \geq 0\}$$

$$w = a^m b^m c^m$$

$$w = uvxyz \quad |vxy| \leq m \quad |vy| \geq 1$$

Case 3: Similar analysis with case 1

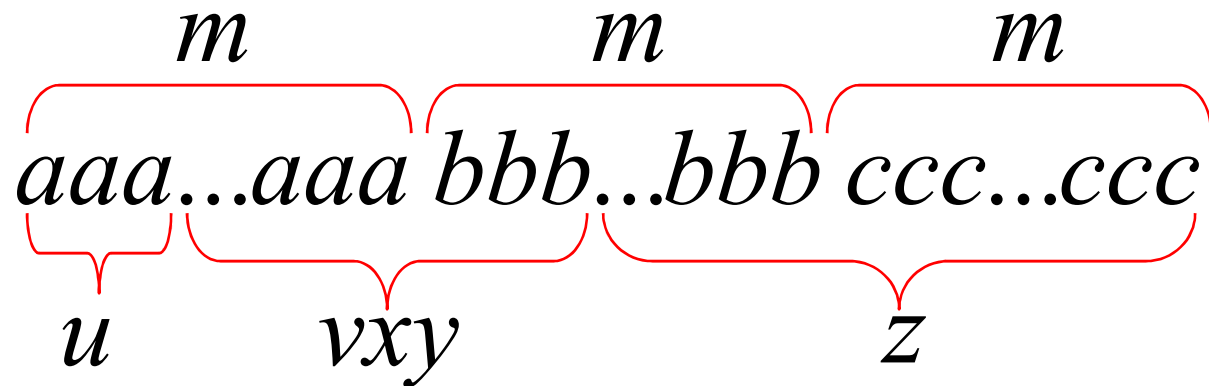
$$\begin{array}{c}
 \overbrace{aaa \dots aaa}^m \quad \overbrace{bbb \dots bbb}^m \quad \overbrace{ccc \dots ccc}^m \\
 \underbrace{aaa \dots aaa \quad bbb \dots bbb}_{u} \quad \underbrace{ccc \dots ccc}_{\begin{array}{c} vxy \quad z \end{array}}
 \end{array}$$

$$L = \{a^n b^n c^n : n \geq 0\}$$

$$w = a^m b^m c^m$$

$$w = uvxyz \quad |vxy| \leq m \quad |vy| \geq 1$$

Case 4: vxy overlaps a^m and b^m

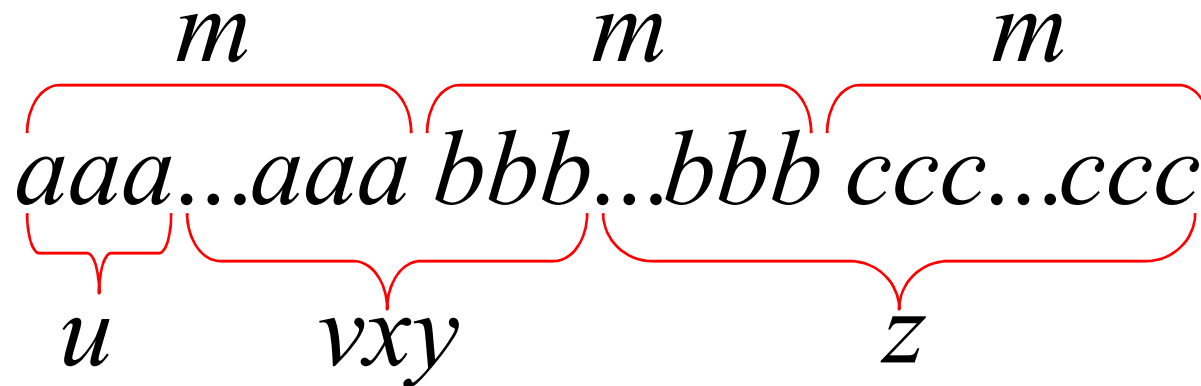


$$L = \{a^n b^n c^n : n \geq 0\}$$

$$w = a^m b^m c^m$$

$$w = uvxyz \quad |vxy| \leq m \quad |vy| \geq 1$$

Case 4: Possibility 1: v contains only a
 y contains only b



$$L = \{a^n b^n c^n : n \geq 0\}$$

$$w = a^m b^m c^m$$

$$w = uvxyz \quad |vxy| \leq m \quad |vy| \geq 1$$

Case 4: Possibility 1: v contains only a

$k_1 + k_2 \geq 1$ y contains only b

$$\begin{array}{c}
 \overbrace{aaa \dots a}^{m+k_1} \quad \overbrace{bbb \dots b}^{m+k_2} \quad \overbrace{ccc \dots c}^m \\
 \underbrace{aaa \dots a}_{u} \quad \underbrace{bbb \dots b}_{v^2 xy^2} \quad \underbrace{ccc \dots c}_z
 \end{array}$$

$$L = \{a^n b^n c^n : n \geq 0\}$$

$$w = a^m b^m c^m$$

$$w = uvxyz \quad |vxy| \leq m \quad |vy| \geq 1$$

Case 4: From Pumping Lemma: $uv^2xy^2z \in L$

$$k_1 + k_2 \geq 1$$

$$\underbrace{\overbrace{aaa \dots a}^{m+k_1}}_u \underbrace{\overbrace{bbb \dots b}^{m+k_2}}_{v^2xy^2} \underbrace{\overbrace{ccc \dots c}^m}_z$$

$$L = \{a^n b^n c^n : n \geq 0\}$$

$$w = a^m b^m c^m$$

$$w = uvxyz \quad |vxy| \leq m \quad |vy| \geq 1$$

Case 4: From Pumping Lemma: $uv^2xy^2z \in L$

$$k_1 + k_2 \geq 1$$

However: $uv^2xy^2z = a^{m+k_1} b^{m+k_2} c^m \notin L$

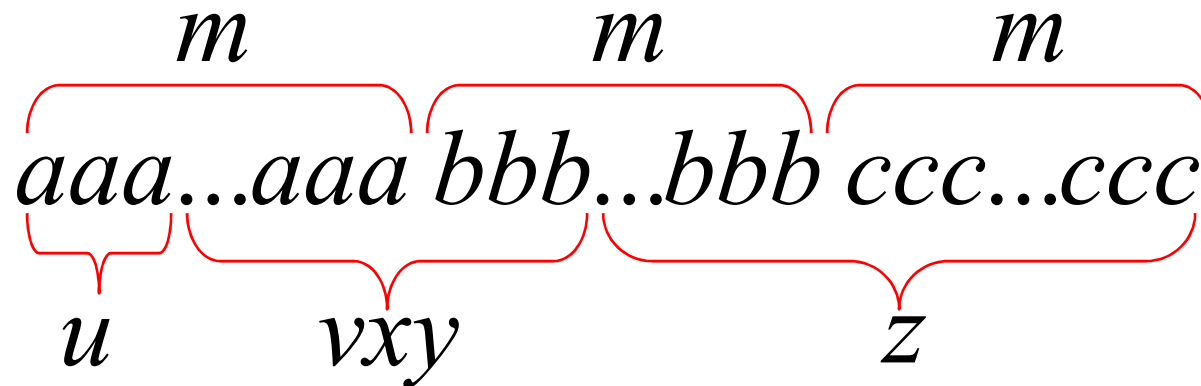
Contradiction!!!

$$L = \{a^n b^n c^n : n \geq 0\}$$

$$w = a^m b^m c^m$$

$$w = uvxyz \quad |vxy| \leq m \quad |vy| \geq 1$$

Case 4: Possibility 2: v contains a and b
 y contains only b



$$L = \{a^n b^n c^n : n \geq 0\}$$

$$w = a^m b^m c^m$$

$$w = uvxyz \quad |vxy| \leq m \quad |vy| \geq 1$$

Case 4: Possibility 2: v contains a and b

$$k_1 + k_2 + k \geq 1 \quad y \text{ contains only } b$$

$$\underbrace{aaa \dots aaa}_{m} \underbrace{abba}_{k_1} \underbrace{abb}_{k_2} \underbrace{bbbbbb \dots bbb}_{m+k} \underbrace{ccc \dots ccc}_{m}$$

$$\underbrace{u}_{aaa \dots aaa} \underbrace{v^2 xy^2}_{abbaabb} \underbrace{z}_{ccc \dots ccc}$$

$$L = \{a^n b^n c^n : n \geq 0\}$$

$$w = a^m b^m c^m$$

$$w = uvxyz \quad |vxy| \leq m \quad |vy| \geq 1$$

Case 4: From Pumping Lemma: $uv^2xy^2z \in L$

$$k_1 + k_2 + k \geq 1$$

$$\underbrace{aaa \dots aaa}_{m} \underbrace{abba}_{k_1} \underbrace{abb}_{k_2} \underbrace{bbbbbb \dots bbb}_{m+k} \underbrace{ccc \dots ccc}_{m}$$

$$\underbrace{u}_{u} \underbrace{v^2 xy^2}_{v^2 xy^2} \underbrace{z}_{z}$$

$$L = \{a^n b^n c^n : n \geq 0\}$$

$$w = a^m b^m c^m$$

$$w = uvxyz \quad |vxy| \leq m \quad |vy| \geq 1$$

Case 4: From Pumping Lemma: $uv^2xy^2z \in L$

However: $k_1 + k_2 + k \geq 1$

$$uv^2xy^2z = a^m b^{k_1} a^{k_2} b^{m+k} c^m \notin L$$

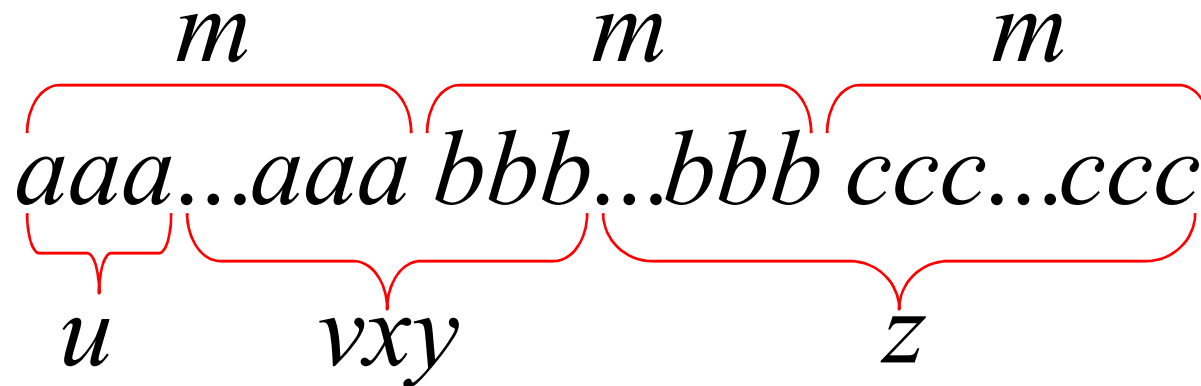
Contradiction!!!

$$L = \{a^n b^n c^n : n \geq 0\}$$

$$w = a^m b^m c^m$$

$$w = uvxyz \quad |vxy| \leq m \quad |vy| \geq 1$$

Case 4: Possibility 3: v contains only a
 y contains a and b



$$L = \{a^n b^n c^n : n \geq 0\}$$

$$w = a^m b^m c^m$$

$$w = uvxyz \quad |vxy| \leq m \quad |vy| \geq 1$$

Case 4: Possibility 3: v contains only a
 y contains a and b

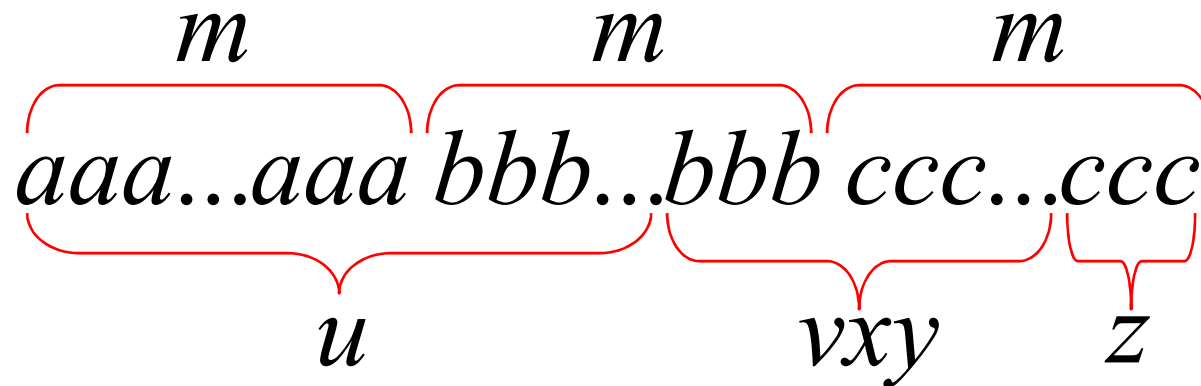
Similar analysis with Possibility 2

$$L = \{a^n b^n c^n : n \geq 0\}$$

$$w = a^m b^m c^m$$

$$w = uvxyz \quad |vxy| \leq m \quad |vy| \geq 1$$

Case 5: vxy overlaps b^m and c^m



$$L = \{a^n b^n c^n : n \geq 0\}$$

$$w = a^m b^m c^m$$

$$w = uvxyz \quad |vxy| \leq m \quad |vy| \geq 1$$

Case 5: Similar analysis with case 4

$$\begin{array}{c}
 \overbrace{aaa \dots aaa}^m \quad \overbrace{bbb \dots bbb}^m \quad \overbrace{ccc \dots ccc}^m \\
 \underbrace{\hspace{1.5cm}}_u \quad \underbrace{\hspace{1.5cm}}_{vxy} \quad \underbrace{\hspace{1.5cm}}_z
 \end{array}$$

There are no other cases to consider

(since $|vxy| \leq m$, string vxy cannot

overlap a^m , b^m and c^m at the same time)

In all cases we obtained a contradiction

Therefore: The original assumption that

$$L = \{a^n b^n c^n : n \geq 0\}$$

is context-free must be wrong

Conclusion: L is not context-free

Prove the Following Theorems

Theorem1: The language

$$L = \{ww : w \in \{a,b\}^*\}$$

is **not** context free

Theorem2: The language

$$L = \{a^{n!} : n \geq 0\}$$

is **not** context free

Theorem3: The language

$$L = \{a^{n^2} b^n : n \geq 0\}$$

is **not** context free